Can a model transcribe French speech?
A documentation of the model

1 Background

To have feedback for participants in an experiment about clarity, having a model to be as the listener would be a logical next step. Instead of training a model from scratch or even taking a speech to text model and fine tuning it, I figured that there has to be a model that was already trained on French speech.

2 Terminology

I used the term fine tuned, what is it actually. Fine tuning a pre-existing model is just taking a model of choice and further training it on a selected dataset. This makes the model be more of an expert in a certain portion of datatypes. So for example, the model I used that would be utilized is the Huang [2022] whisper-medium model for ASR (automatic speech recognition) in French model which uses OpenAI's medium sized whisper model [Radford et al., 2022, 2023]. The French ASR has been trained "on a composite dataset comprising of over 2200 hours of French speech audio, using the train and the validation splits of Common Voice 11.0, Multilingual LibriSpeech, Voxpopuli, Fleurs, Multilingual TEDx, MediaSpeech, and African Accented French" [Ardila et al., 2020, Pratap et al., 2020, Wang et al., 2021, Conneau et al., 2022, Salesky et al., 2021, Kolobov et al., 2021, gigant, 2022].

ASR models are a type of model that does speech to text transcription. ASR models "converts a speech signal to text, mapping a sequence of audio inputs to text outputs" [Hugging Face, 2024b].

3 Methodology

As mentioned in section 2, I used the whisper-medium model for ASR in French [Huang, 2022]. This is already a powerful model by itself with the amount of hours it was trained on. The model has 769 million parameters to tweak which will come in handy later on.

3.1 Data

Since we have data that was already collected and have the actual transcription, I decided to fine tune the model further with the data. To make sure that the model actually learns to not include any of the annotation anomolies like $\{$np$\}$, $\{$cough$\}$, $\{$sp$\}$, etc, I made sure to remove these. I noticed that if there was some discrempacies with ce and le/la when comes before a vowel so I removed the space that would have been put when extracting from the textgrid. The data can be seen here: lookitsluc1 [2024]. I split the data 80% to 20%. The training split has around 4.59 thousand elements and the test split has around 1.15 thousand elements.

3.2 Fine tuned training

3.2.1 Setup

To make the model ready to be fine-tuned, there are some elements that were added on top to get the metrics to a good point. Since whisper-medium-french [Huang, 2022] has 769 million parameters, I did not want to tweak the parameters that gives the power of the whisper model to do the speech-to-text task. To only target the parameters that would best to tweak in my favor, I added a PEFT (Parameter-Efficient Fine-Tuning) via LoRA (Low-Rank Adaptation) wrapper on top of the base model (whisper-medium-french [Huang, 2022]).

LoRA freezes base model's weights and adds an adapter on top that runs along side model and is task specific. PEFT is a way to lower computation times and tunes a small subset of parameters for the task. PEFT takes additional parameters and changes and tweaks them. Adds small, trainable modules or updates a limited set of parameters instead of the full model; Learns task-specific patterns without affecting the core pre-trained model; Maintains efficiency while enabling easy customization using techniques like adapters, LoRA, and prompt tuning [GeeksforGeeks, 2023].

To setup the LoRA and PEFT for the whisper-medium-french [Huang, 2022], there are some parameters I explicitly set. The first is rank (r), which is the amount of parameters to be trained so ”a higher rank means the model has more parameters to train, but it also means the model has more learning capacity,” [Hugging Face, 2024a]. In the model I have set the rank to 8. The lora_alpha is the scaling factor which I put as 16. The bias is “whether none, all or only the LoRA bias parameters should be trained” [Hugging Face, 2024a] which I set to “none”.

3.2.2 Worry about fine-tuning the model

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